

AYUSH BAJPAI

mrayushbajpai.github.io

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PROFESSIONAL SUMMARY

Entry-level Software Engineer with strong foundations in **computer science fundamentals**, **data structures**, **object-oriented programming**, and **software development lifecycle (SDLC)**. Hands-on experience **building and maintaining production-ready applications**, writing clean and documented code, and working with **version control**, **testing**, **debugging**, **APIs**, and **databases**. Demonstrated ability to learn quickly, collaborate with cross-functional teams, and deliver features within defined scope and timelines.

SKILLS

Programming Languages: C, C++, Java, Python, HTML, CSS, SQL

Development Tools: Git, GitHub, Postman, CMake, Ninja, MySQL, Slack, Linux, REST APIs, Webhooks

Testing and Debugging: Unit Testing (basic), Debugging, Exception Handling, Logging

Development Practices: Version Control, Code Reviews, Documentation, Agile/Scrum Fundamentals

CS Fundamentals: Data Structures & Algorithms (DSA), Operating System (OS), Object Oriented Programming (OOP), Database Management System (DBMS), Software Development Lifecycle (SDLC)

PROJECTS EXPERIENCE

Auto Brightness Windows

[\(GitHub | Download Link\)](#)

Python, OpenCV, Numpy, Tkinter, Pystray, Threading, Color Theory

- Designed and deployed a **Python-based script** and executable to enable **adaptive brightness on Windows** systems **lacking native support**.
- Leveraged **OpenCV for camera input** and real-time ambient light estimation, reducing manual brightness adjustments by **~90%** and enhancing user comfort.

Aethern - Adaptive Spaced Repetition Scheduler (SM-2)

[\(GitHub\)](#)

C++, CMake, Ninja, vcpkg, SM-2 Algorithm, Cryptography, Data Structures, System Design

- Designed a **secure, persistent spaced-repetition scheduler** using the **SM-2 algorithm** for optimized long-term memory retention.
- Implemented **AES-256 encryption** with Argon2 password hashing to protect sensitive user data.
- Engineered a **scalable C++ architecture** using **efficient data structures** (hash maps, vectors) with zero plaintext persistence.
- Implemented **end-to-end encrypted local storage** using **XSalsa20-Poly1305 (AEAD)** for item payloads and metadata, ensuring **confidentiality, integrity, and authenticity** of stored records.

IoT-Based Air Quality Monitoring System

[\(Link\)](#)

ESP32, MQ135, DHT11, Arduino IoT Cloud, C++

- Designed and developed an **IoT-enabled** air quality monitoring system to measure **AQI, Temperature**, and **humidity** in real time.
- Built a responsive **Android Application** for **visualizing live sensor data**, enhancing accessibility across devices
- Implemented **Arduino IoT Cloud** for **real-time data synchronization**, reducing latency by **~20%** and ensuring **24/7 system uptime**.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

2022-2026

G.L. Bajaj Institute of Technology and Management

Current CGPA - 7.74

Class XII (CBSE) - 85.6%

2021-2022

Class X (CBSE) - 85.6%

2019-2020

CO-CURRICULAR & ACHIEVEMENT

- Design & Media Team Lead**, departmental club - delivered **50+ digital assets** at a **100% on time delivery**, demonstrating leadership, ownership and accountability.
- Led 10+ robotics events** from **hosting to execution**, overseeing **planning, scheduling** and **design**.
- Coached a Semi Finalist, National-Level India College Chess Championship** hosted by **chess.com**, demonstration teamwork, mentorship and analytical thinking.
- Rated **1850+** on **Leetcode Contests** as a **Leetcode Knight**